

Adriza Mishra

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Technical Skills

AI / ML: Agentic RAG, LLM Fine-tuning (Vertex AI Gemini), Embeddings, Anomaly Detection, MLFlow, TensorFlow, Keras
Backend & Frameworks: FastAPI, Spring Boot, Async Microservices, Pydantic, REST APIs, CI/CD
Infrastructure: Kubernetes, Docker, Helm, Databricks, Azure Cloud, GCP, PostgreSQL, Redis, GitHub Actions
Languages: Python, Java, TypeScript

Education

Indian Institute of Technology Madras
Dual Degree in Biotechnology

July 2018 – June 2023
CGPA: 8.02

Professional Experience

Honeywell

July 2023 – Present

Software Engineer II

AI-Driven Rule Translation Pipeline | AXON → DSL | Production

- Designed an **agentic RAG pipeline** replacing a legacy SkySpark dependency; translates 500+ building energy management rules from AXON (Project Haystack) to proprietary DSL with ANTLR grammar constraints
- Architected **layered retrieval** across 4 separate indexes (syntax constraints, rule-family templates, utility patterns, gold examples), reducing LLM hallucination of reserved keywords and invalid constructs
- Built a **DSL syntax validator** as post-generation oracle — auto-accepts valid translations, routes failures to context-aware retry or human review; measurably reduced manual review vs. pure prompt engineering baseline
- Integrated **fine-tuned Vertex AI Gemini** (JSONL/GCP) on top of RAG baseline; correction feedback loops back into the knowledge base to improve future translations
- Shipped as **FastAPI async service** with PostgreSQL-backed versioned prompt and system-instruction management; full per-request tracking of failures and corrections; **eliminated SkySpark vendor dependency** entirely

ML Training Orchestration & Inference Platform

- Engineered **ML training orchestration** on Databricks (Python, MLFlow, PySpark) for 100+ models across 20+ multi-tenant clients; MLflow tracked experiments, versioned artifacts, and managed lifecycle transitions (Staging → Production → Archived) — reduced manual setup 80%, maintained 99.5% job success rate
- Built **FastAPI async services** for model onboarding, inference, and deployment – Pydantic schemas and dependency injection handled concurrent multi-tenant requests; inference layer processed 10K+ daily predictions across 50+ models at sub-100ms latency; async Delta Lake writes and drift detection cut debugging cycles 60%
- Developed **containerized deployment pipeline** dispatching models to Kubernetes or Databricks; cut deploy time from 3 days to 30 minutes, infra costs 35% via optimized container sharing
- Implemented **autoencoder anomaly detection** for industrial IoT with drift monitoring across 50+ variables; extended to 10+ customer model types (TensorFlow, PyTorch, scikit-learn)

Security, Quality & LLM Testing

- Designed a security validation pipeline across 5 tools (Coverity, ModelScan, Twistlock, Black Duck Hub, Malware Scan); 8+ automated checks cut validation time 75%, caught 40+ vulnerabilities, maintained 100% SOC 2 compliance
- Won company-wide bug bash** on Honeywell's flagship LLM assistant; authored LLM testing whitepaper, developed proprietary frameworks yielding multiple patent submissions; automated 150+ test scenarios (TypeScript WebDriverIO + Python Behave), compressing regression from 40hrs to 5hrs/sprint

Tools: MLFlow, FastAPI, Databricks, Unity Catalog, TensorFlow, Vertex AI, GCP, Coverity, ModelScan, Twistlock, Black Duck Hub, Azure Cloud, PostgreSQL, Azure Delta Lake, Kubernetes, Python Behave, TypeScript, WebDriverIO

Portfolio

Lox Interpreter & Playground API

February 2024 – Present

- Implemented a **tree-walk interpreter** for Lox in Java – full pipeline from lexical analysis through recursive descent parsing to runtime evaluation; closures, lexical scoping, classes, inheritance, and optimized scope resolution via environment chaining
- Extended with custom data types, color-coded error diagnostics, and call stack management for nested closures; exposed as **Spring Boot 3 REST API** (Maven multi-module) with wall-clock timeout, Bean Validation, and integration test suite

Tools: Java 17, Spring Boot 3, Maven, JUnit, Recursive Descent Parsing, Visitor Pattern, Environment Chaining

HylyTool — Chrome Extension | Study Tool

May 2024 – July 2024

- Built TypeScript/Manifest V3 extension with **three-tier DOM text-matching algorithm**; 100% highlight accuracy on reload with dynamic content via TreeWalker, MutationObserver, Range/Selection APIs
- Persistent highlights across sessions and devices via Chrome Storage API with URL-based keying; real-time cross-tab manipulation via message passing between extension components
- Built **resilient highlight recovery** to survive DOM mutations and dynamic content reloads; added **per-highlight notes** anchored to selections, **one-click export** of annotated highlights, and **Chrome account sync** for cross-device persistence when signed into Chrome

Tools: TypeScript, Webpack, Chrome Manifest V3, Chrome Storage API, DOM APIs